Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
Fall 2026

Recitation 11 Handout

Tuesday, October 20

Problem 11.1

Let $X \in \{1, 2, \ldots\}$ be a random variable that represents the number of coin tosses until we observe the first heads. Suppose we repeat the experiment n times, generating values of $X_1, \ldots, X_n$ that are independent, identically distributed geometric random variables.

(a) We use the familiar parameterization of the geometric distribution

$$p_{X_i}(x; p) = (1-p)^{x-1}\, p, \qquad x = 1, 2, \ldots,$$

where $p \in (0, 1)$ is the probability of heads. Determine the maximum likelihood (ML) estimate $\hat{p}_{\text{ML}}$ of the parameter p based on the observed values $x_1^n \triangleq (x_1, \ldots, x_n)$.

(b) Now we use an alternative parameterization

$$p_{X_i}(x; \theta) = \left(1 - \frac{1}{\theta}\right)^{x-1} \frac{1}{\theta}, \qquad x = 1, 2, \ldots,$$

where $\theta \in (1, \infty)$ is the expectation of X, $\theta \triangleq \mathbb{E}[X_i] = 1/p$. Determine the ML estimate $\hat{\theta}_{\text{ML}}$ of the parameter θ based on the observed values x_1^n.

(c) Determine bias $b_{\hat{\theta}_{\text{ML}}} \triangleq \mathbb{E}\Big[\hat{\theta}_{\text{ML}}(X_1, \ldots, X_n)\Big] - \theta$ and variance $\text{var}\Big(\hat{\theta}_{\text{ML}}(X_1, \ldots, X_n)\Big)$ of the ML estimator you found in part (b).

Problem 11.2

Suppose a vendor produced N widgets, numbered $1, 2, \ldots, N$. Its competitor observes $n \leq N$ widgets with numbers $X_1, X_2, \ldots, X_n$. Assume that these numbers are drawn uniformly without replacement from $\{1, 2, \cdots, N\}$.

(a) Determine the ML estimate $\hat{N}_{\text{ML}}$ of N based on the observed widgets $X_1 = x_1, \ldots, X_n = x_n$.

(b) Determine the bias of the ML estimator $b_{\hat{N}_{\text{ML}}} = \mathbb{E}\Big[\hat{N}_{\text{ML}}\Big] - N$. You may find this fact helpful:

$$\sum_{m=n}^{N} m \binom{m-1}{n-1} = n \binom{N+1}{n+1}.$$